The Standard Model on the world brane of 7+1 spacetime, the portals of dirac16complex, and neutrino flavour

The complete Standard Model in the gravitational field of 7+1 spacetime, electroweak symmetry breaking, quark and lepton masses and mixing, dirac16complex at the brane as four Dirac spinors, Kaluza–Klein masses, the portals and the coupled field equations, and neutrino flavour conversion in vacuum, in matter, in the Sun and in supernovae — sections 17–19 of the notebook 7+1.nb

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Abstract

The 7+1 spacetime of this project has one time coordinate x_0, three ordinary space coordinates x_1, x_2, x_3 and four auxiliary space coordinates $x_4, \ldots, x_7$. This document places the complete Standard Model of particle physics on the 3+1-dimensional WORLD BRANE $x_4 = x_5 = x_6 = x_7 = 0$, in the gravitational field of the whole spacetime, and couples it to the 16-component field dirac16complex, which lives in the whole spacetime. It states the field content and the Lagrangian, checks anomaly cancellation and electroweak symmetry breaking with current data, decomposes dirac16complex at the brane into four ordinary Dirac spinors, derives its Kaluza–Klein masses, defines three portals and the coupled field equations, and works out neutrino flavour conversion with and without dirac16complex. Sections 17–19 of 7+1.nb contain 23 checks (23 True); the independent Python program tools/ref_sm.py contains 32 (32 True).

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1. Results at a glance

- **Field content.** Gauge group $SU(3)_C \times SU(2)_L \times U(1)_Y$, three generations including right-handed neutrinos, one Higgs doublet; all gauge anomalies cancel in every generation (section 3).
- **Electroweak scale.** $v = (\sqrt{2}G_F)^{-1/2} = 246.2196402$ GeV; tree-level $m_W = gv/2$, $m_Z = \sqrt{g^2+g'^2}\,v/2$, $m_H^2 = 2\lambda v^2$ (section 5).
- **Mixing.** CKM matrix with the Jarlskog invariant $J = 3.117 \times 10^{-5}$; PMNS matrix from NuFit-6.0 (section 6).
- **Cosmology on the brane.** The brane Friedmann equation gives the age 13.791 Gyr (section 7).
- **dirac16complex at the brane** is four brane Dirac spinors; a mode with auxiliary momentum $\vec{q}$ has the Kaluza–Klein mass $\sqrt{m^2 + |\vec{q}|^2/b_0^2}$ (sections 8–9).
- **Portals** (Yukawa, vector, Higgs) couple dirac16complex to the Standard Model at the brane; the Yukawa portal makes the light neutrino mixing non-unitary (section 10).
- **Neutrino flavour.** At DUNE ($L = 1300$ km, $E = 2.5$ GeV) $P(\nu_\mu \to \nu_e) = 0.048837$ in vacuum and 0.072327 in matter (section 11).

2. The world brane and the total action

Units $\hbar = c = 1$. The auxiliary coordinates form a torus T^4: $x_k \simeq x_k + L_c$ ($k = 4, \dots, 7$; coordinate period L_c, proper size bL_c). The Standard Model lives on the world brane

$$W = \{x_4 = x_5 = x_6 = x_7 = 0\}, \qquad x = (x_0, x_1, x_2, x_3), \quad y = (x_4, \dots, x_7),$$
$$h_{\mu\nu}(x) = g_{\mu\nu}(x, y = 0) \quad (\mu, \nu = 0, \dots, 3),$$

with the induced metric $h_{\mu\nu}$; gravity and dirac16complex live in the whole 7+1 spacetime (the bulk). The total action is

$$S = S_{\mathrm{EL}}[g] + S_\psi[\psi, g] + \int_W d^4x\,\sqrt{|h|}\,(\mathcal{L}_{\mathrm{SM}} + \mathcal{L}_{\mathrm{portal}}) + S_{\mathrm{stab}}, \qquad \int_W d^4x\,\sqrt{|h|}\,(\dots) = \int d^8X\,\sqrt{|g|}\,\frac{\delta^4(y)}{b^4}\,(\dots),$$

where the last identity holds for the cosmological metric $ds^2 = dx_0^2 - a^2\sum_{i=1}^3 dx_i^2 - b^2\sum_{k=4}^7 dx_k^2$ ($\sqrt{|g|} = a^3b^4$, $\sqrt{|h|} = a^3$). The pieces:

$$S_{\mathrm{EL}} = -\frac{1}{2\kappa}\int d^8X\sqrt{|g|}\sum_{k=0}^{3}\alpha_k L_k, \qquad L_k = \frac{1}{2^k}\delta^{\alpha_1\beta_1\cdots\alpha_k\beta_k}_{\gamma_1\delta_1\cdots\gamma_k\delta_k} R^{\gamma_1\delta_1}{}_{\alpha_1\beta_1}\cdots R^{\gamma_k\delta_k}{}_{\alpha_k\beta_k},$$

$$S_\psi = \int d^8X\sqrt{|g|}\Big[\bar\psi\big(i\Gamma^\mu(X)D_\mu - m\big)\psi - \frac{1}{2}\sum_{r=0}^{8} g_r \sum_{|K|=r}(\bar\psi\Theta_K\psi)(\bar\psi\Theta^K\psi)\Big],$$

($\kappa = 8\pi G_8$; $\alpha_0 = 2\bar\Lambda$, $\alpha_1 = 1$; $\Gamma^\mu(X)$ the generalized frame matrices, $D_\mu = \partial_\mu + \Omega_\mu$ with the canonical spin connection Ω_μ; Θ^K the 256 Clifford products with g_r the nine self-couplings of dirac16complex), and S_{stab} is a stress that keeps b constant in the modern epoch.

3. Fields, quantum numbers and anomaly cancellation (section 17.1)

Gauge fields G^a_μ ($a = 1,\dots,8$), W^I_μ ($I = 1,2,3$), B_μ with couplings g_s, g, g'. Matter in each of three generations (Weyl fields; electric charge $Q = T_3 + Y$):

field	$SU(3)_C$	$SU(2)_L$	Y	**chirality** χ_f	**charges**
$Q_L = (u_L, d_L)$	3	2	1/6	+1 (left)	2/3, −1/3
u_R	3	1	2/3	−1 (right)	2/3
d_R	3	1	−1/3	−1	−1/3
$L_L = (\nu_L, e_L)$	1	2	−1/2	+1	0, −1
e_R	1	1	−1	−1	−1
ν_R	1	1	0	−1	0
Φ (Higgs)	1	2	1/2	—	$\tilde\Phi = i\sigma_2\Phi^*$

With N_f the number of colours and d_f the $SU(2)$ dimension of field f, the anomaly coefficients of one generation are

$$A_{SU(3)^2U(1)} = \sum_{f:\,N_f=3} \chi_f d_f Y_f, \qquad A_{SU(2)^2U(1)} = \sum_{f:\,d_f=2} \chi_f N_f Y_f, \qquad A_{SU(3)^3} = \sum_{f:\,N_f=3} \chi_f d_f,$$
$$A_{U(1)^3} = \sum_f \chi_f N_f d_f Y_f^3, \qquad A_{\mathrm{grav}^2\mathrm{U}(1)} = \sum_f \chi_f N_f d_f Y_f,$$

and all five vanish, with or without ν_R; the number of $SU(2)$ doublets per generation ($3+1=4$) is even (no Witten anomaly). The relativistic degrees of freedom above the electroweak scale (without ν_R) are

$$g_* = \underbrace{16}_{G} + \underbrace{6}_{W} + \underbrace{2}_{B} + \underbrace{4}_{\Phi} + \frac{7}{8}\cdot 3\cdot\big(12+6+6+4+2\big) = 28 + \frac{7}{8}\cdot 90 = 106.75.$$

4. The Lagrangian on the brane, in the gravitational field (section 17.2)

With the frame (tetrad) of $h_{\mu\nu}$, its spin connection and complex 4×4 Dirac matrices $\gamma^\mu(x)$ of the brane ($\gamma^\mu\gamma^\nu + \gamma^\nu\gamma^\mu = 2h^{\mu\nu}$), and the gauge-covariant derivative

$$D_\mu = \nabla_\mu - ig_s\, G^a_\mu T^a - ig\, W^I_\mu \frac{\tau^I}{2} - ig'\, Y\, B_\mu$$

(T^a the generators of $SU(3)$ in the representation of the field, τ^I the Pauli matrices; ∇_μ includes the spin connection of h on spinors), the Standard Model Lagrangian is

$$\mathcal{L}_{\mathrm{SM}} = -\tfrac14 G^a_{\mu\nu}G^{a\mu\nu} - \tfrac14 W^I_{\mu\nu}W^{I\mu\nu} - \tfrac14 B_{\mu\nu}B^{\mu\nu} + (D_\mu\Phi)^\dagger(D^\mu\Phi) - \mu^2\Phi^\dagger\Phi - \lambda(\Phi^\dagger\Phi)^2$$
$$+ \sum_f \bar f\, i\gamma^\mu D_\mu f - \big[\bar Q_L Y_u \tilde\Phi u_R + \bar Q_L Y_d \Phi d_R + \bar L_L Y_e \Phi e_R + \bar L_L Y_\nu \tilde\Phi \nu_R + \mathrm{h.c.}\big],$$
$$G^a_{\mu\nu} = \partial_\mu G^a_\nu - \partial_\nu G^a_\mu + g_s f^{abc} G^b_\mu G^c_\nu, \quad W^I_{\mu\nu} = \partial_\mu W^I_\nu - \partial_\nu W^I_\mu + g\,\epsilon^{IJK} W^J_\mu W^K_\nu, \quad B_{\mu\nu} = \partial_\mu B_\nu - \partial_\nu B_\mu,$$

f runs over all Weyl fields of the three generations, and Y_u, Y_d, Y_e, Y_ν are complex 3×3 Yukawa matrices. Its field equations are the ordinary ones with $\partial \to \nabla$ (the Levi-Civita connection of h), e.g. $\nabla_\mu G^{a\mu\nu} + g_s f^{abc} G^b_\mu G^{c\mu\nu} = -g_s \sum_q \bar q \gamma^\nu T^a q$.

5. Electroweak symmetry breaking (section 17.3)

With $\langle\Phi\rangle = (0, v/\sqrt{2})$ the term $(D_\mu\langle\Phi\rangle)^\dagger(D^\mu\langle\Phi\rangle)$ is a mass matrix for (W^1, W^2, W^3, B), its Hessian $M^2_{ij} = \partial^2\big|g\,W^I\tfrac{\tau^I}{2}\langle\Phi\rangle + \tfrac{g'}{2}B\langle\Phi\rangle\big|^2/\partial w_i\partial w_j$ having the eigenvalues

$$m_W^2 = \frac{g^2v^2}{4}\ \text{(twice)}, \qquad m_Z^2 = \frac{(g^2+g'^2)v^2}{4}, \qquad m_\gamma^2 = 0.$$

Inputs (Particle Data Group 2025): $G_F = 1.1663788\times10^{-5}$ GeV^{-2}, $m_W = 80.3692$ GeV, $m_Z = 91.1880$ GeV, $m_H = 125.20$ GeV. At tree level (on-shell weak angle)

$$v = (\sqrt{2}G_F)^{-1/2}, \quad g = \frac{2m_W}{v}, \quad \cos\theta_W = \frac{m_W}{m_Z}, \quad g' = g\tan\theta_W, \quad e = g\sin\theta_W, \quad \lambda = \frac{m_H^2}{2v^2},$$

v **(GeV)**	g	g'	$\sin^2\theta_W$	e	$1/\alpha = 4\pi/e^2$	λ
246.2196402	0.6528252573	0.3499462340	0.2232094928	0.3084276316	132.1001486	0.1292805757

($1/\alpha$ here is the tree-level value at the weak scale, not the low-energy fine-structure constant.)

6. Quark and lepton masses; CKM and PMNS mixing (section 17.4)

Fermion masses are $m = yv/\sqrt{2}$ for the singular values y of the Yukawa matrices; the mixing matrices are $V_{\rm CKM} = U_{uL}^\dagger U_{dL}$ and $U_{\rm PMNS} = U_{eL}^\dagger U_{\nu L}$, where $Y = U_L\,\mathrm{diag}(y)\,U_R^\dagger$. Both are written in the standard parametrization ($c_{ij} = \sqrt{1-s_{ij}^2}$):

$$U(s_{12}, s_{13}, s_{23}, \delta) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & c_{23} & s_{23} \\ 0 & -s_{23} & c_{23} \end{pmatrix} \begin{pmatrix} c_{13} & 0 & s_{13}e^{-i\delta} \\ 0 & 1 & 0 \\ -s_{13}e^{i\delta} & 0 & c_{13} \end{pmatrix} \begin{pmatrix} c_{12} & s_{12} & 0 \\ -s_{12} & c_{12} & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Quarks (PDG 2024 CKM fit: $s_{12} = 0.22501$, $s_{13} = 0.003732$, $s_{23} = 0.04183$, $\delta = 1.147$; PDG 2025 masses: $m_u = 2.16$ MeV, $m_d = 4.70$ MeV, $m_s = 93.5$ MeV, $m_c = 1.2730$ GeV, $m_b = 4.183$ GeV, $m_t = 172.56$ GeV). With $Y_u = \sqrt{2}\,\mathrm{diag}(m_u, m_c, m_t)/v$ and $Y_d = \sqrt{2}\,V\,\mathrm{diag}(m_d, m_s, m_b)/v$ the singular values reproduce the masses and $|U_{uL}^\dagger U_{dL}| = |V|$. The notebook obtains

$$|V_{\rm CKM}| = \begin{pmatrix} 0.974350 & 0.225008 & 0.00373200 \\ 0.224876 & 0.973489 & 0.0418297 \\ 0.00858280 & 0.0411095 & 0.999118 \end{pmatrix},$$

$$J = \mathrm{Im}\big(V_{us}V_{cb}V_{ub}^*V_{cs}^*\big) = 3.11699\times10^{-5}, \qquad \bar\rho + i\bar\eta = -\frac{V_{ud}V_{ub}^*}{V_{cd}V_{cb}^*} = 0.159184 + 0.352275\,i,$$

$$A = \frac{s_{23}}{s_{12}^2} = 0.826198, \qquad y_t = \frac{\sqrt{2}m_t}{v} = 0.99113414,$$

in agreement with the PDG 2024 values $J = (3.12^{+0.13}_{-0.12})\times10^{-5}$, $A = 0.826$, $\bar\rho = 0.1591$, $\bar\eta = 0.3523$.

Leptons (NuFit-6.0, normal ordering, without the atmospheric data of Super-Kamiokande: $\sin^2\theta_{12} = 0.307$, $\sin^2\theta_{23} = 0.561$, $\sin^2\theta_{13} = 0.02195$, $\delta = 177°$, $\Delta m^2_{21} = 7.49\times10^{-5}$ eV², $\Delta m^2_{31} = 2.534\times10^{-3}$ eV², $\Delta m^2_{ij} = m_i^2 - m_j^2$). With ν_R the neutrinos are Dirac particles; with the lightest mass $m_1 = 0$: $m_2 = \sqrt{\Delta m^2_{21}}$, $m_3 = \sqrt{\Delta m^2_{31}}$, and $Y_\nu = \sqrt{2}\,U\,\mathrm{diag}(m_1, m_2, m_3)/v$ (electron Yukawa basis diagonal):

$$|U_{\rm PMNS}| = \begin{pmatrix} 0.82328 & 0.54796 & 0.14816 \\ 0.27491 & 0.61298 & 0.74073 \\ 0.49663 & 0.56921 & 0.65526 \end{pmatrix}, \qquad J_{\rm PMNS} = 0.0017359,$$

$$(m_1, m_2, m_3) = (0, 0.00865448, 0.0503389)\ \text{eV}.$$

7. The modern epoch on the brane (section 17.5)

With $b = b_0$ the 00 field equation is the Friedmann equation of the brane,

$$3H_a^2 = 8\pi G_4\,\rho_4 + \Lambda_4, \qquad G_4 = \frac{G_8}{V_\perp}, \qquad V_\perp = b_0^4 L_c^4, \qquad \varepsilon_8 = \frac{\rho_4}{V_\perp} \;\; \text{(matter on the brane)},$$

($H_a = a'/a$, ρ_4 the four-dimensional energy density). With the Planck 2018 parameters $H_0 = 67.4$ km/s/Mpc, $\Omega_m = 0.315$, $\Omega_r = 9.14 \times 10^{-5}$, $\Omega_\Lambda = 1 - \Omega_m - \Omega_r$ the age of the universe is

$$t_0 = \frac{1}{H_0}\int_0^1 \frac{da}{a\sqrt{\Omega_r a^{-4} + \Omega_m a^{-3} + \Omega_\Lambda}} = 13.7907 \text{ Gyr} \qquad \text{(Planck 2018: } 13.787 \pm 0.020\text{)}.$$

With the reduced eight-dimensional Planck mass M_8 ($8\pi G_8 = M_8^{-6}$) the reduced four-dimensional one is $M_{\rm Pl}^2 = M_8^6 V_\perp$ ($M_{\rm Pl} = 2.435 \times 10^{18}$ GeV). For $M_8 = 10$ TeV:

$$V_\perp = \frac{M_{\rm Pl}^2}{M_8^6} = 5.92923 \times 10^{12}\ \text{GeV}^{-4}, \qquad V_\perp^{1/4} = 3.07919 \times 10^{-13}\ \text{m}, \qquad \frac{2\pi}{V_\perp^{1/4}} = 4.02652\ \text{MeV},$$

the last being the lowest Kaluza–Klein mass of a field on the auxiliary torus (proper period $V_\perp^{1/4} = b_0 L_c$).

8. dirac16complex at the brane: four brane Dirac spinors (section 18)

At the brane the eight Dirac matrices split into four along W and four transverse to it. With the complex 4×4 Dirac matrices of the Dirac representation and four anti-Hermitian 4×4 matrices ρ_k,

$$\gamma_0 = \mathrm{diag}(1,1,-1,-1), \quad \gamma_i = \begin{pmatrix} 0 & \sigma_i \\ -\sigma_i & 0 \end{pmatrix}, \quad \gamma_5 = i\gamma_0\gamma_1\gamma_2\gamma_3, \qquad \gamma_\mu\gamma_\nu + \gamma_\nu\gamma_\mu = 2\eta_{\mu\nu}1_4,$$
$$\rho_4 = i\gamma_0, \quad \rho_5 = -\gamma_1, \quad \rho_6 = -\gamma_2, \quad \rho_7 = -\gamma_3, \qquad \rho_k\rho_l + \rho_l\rho_k = -2\delta_{kl}1_4,$$

(σ_i the Pauli matrices, $\eta = \mathrm{diag}(1,-1,-1,-1)$) the sixteen-dimensional matrices

$$R_\mu = \gamma_\mu \otimes 1_4 \;\; (\mu = 0,\ldots,3), \qquad R_k = \gamma_5 \otimes \rho_k \;\; (k = 4,\ldots,7)$$

satisfy the same Clifford relations as the real Dirac matrices Γ_A of 7+1 spacetime, $R_A R_B + R_B R_A = 2\eta_{AB}1_{16}$ with $\eta = \mathrm{diag}(+1,-1,\ldots,-1)$. Both representations are irreducible, so there is a unitary U with $\Gamma_A U = U R_A$; it is obtained as the Schur average over the 256 Clifford products,

$$U \propto \sum_K \Gamma^{(K)}\, s\, \big(R^{(K)}\big)^{-1}, \qquad \Gamma^{(K)} = \Gamma_{A_1}\cdots\Gamma_{A_r}, \quad R^{(K)} = R_{A_1}\cdots R_{A_r},$$

with a fixed matrix s (the first matrix unit for which the sum is non-zero), normalized so that $U^\dagger U = 1_{16}$. Writing $\psi = U\zeta$ with $\zeta \in \mathbb{C}^4 \otimes \mathbb{C}^4$, $\zeta = \sum_{a=1}^4 \psi_a \otimes e_a$, dirac16complex at the brane is FOUR brane Dirac spinors ψ_a (a the transverse spinor index), and with $\bar\psi_a = \psi_a^\dagger\gamma_0$, $\gamma^\mu = \eta^{\mu\mu}\gamma_\mu$:

$$\bar\psi\psi' = \sum_{a=1}^4 \bar\psi_a\psi'_a, \qquad \bar\psi\,\Gamma^\mu\psi' = \sum_{a=1}^4 \bar\psi_a\gamma^\mu\psi'_a \;\; (\mu = 0,\ldots,3).$$

The eight-dimensional chirality $\chi = i\Gamma_0\cdots\Gamma_7$ is

$$\chi = U\,(\gamma_5 \otimes \rho_{4567})\,U^\dagger, \qquad \rho_{4567} = \rho_4\rho_5\rho_6\rho_7 \;\; \text{(Hermitian, eigenvalues } +1,+1,-1,-1),$$

so the left-handed half ψ_L ($\chi = -1$) contains brane-right-handed parts with $\rho_{4567} = -1$ and brane-left-handed parts with $\rho_{4567} = +1$.

9. Kaluza–Klein masses on the auxiliary torus (section 18.1)

With $b = b_0$ and $\psi \propto e^{i\vec{q}\cdot\vec{y}}$, $\vec{q} = 2\pi\vec{n}/L_c$ ($\vec{n} \in \mathbb{Z}^4$), the eight-dimensional Dirac equation at the brane becomes, in the variables ζ,

$$\big(\gamma\cdot p\otimes 1_4 + X - m\big)\zeta = 0, \qquad \gamma\cdot p = p_0\gamma_0 - \sum_{i=1}^{3} p_i\gamma_i, \qquad X = \gamma_5 \otimes \frac{\vec{\rho}\cdot\vec{q}}{b_0}, \qquad X^2 = -\frac{|\vec{q}|^2}{b_0^2}1_{16},$$

and X anticommutes with every $\gamma_\mu \otimes 1_4$. Hence each Fourier mode is four brane Dirac fermions of mass

$$m_{\vec{n}} = \sqrt{m^2 + \frac{|\vec{q}|^2}{b_0^2}} \qquad \text{(the zero mode } \vec{n} = 0 \text{ has mass } m),$$

checked by the vanishing of $\det(\gamma\cdot p\otimes 1_4 + X - m)$ on $p_0^2 - |\vec{p}|^2 = m_{\vec{n}}^2$ (and not off shell), in 40-digit arithmetic, for $m = 7/10$, $b_0 = 13/10$, $\vec{q} = 2\pi(2, -1, \frac{1}{2}, 3)$, $\vec{p} = (0.3, -0.2, 0.4)$.

10. The portals and the coupled field equations (section 18.2)

The portals of dirac16complex are local couplings at the brane (an invented interaction; separate couplings for the two eight-dimensional chiral halves), with $\psi_a(x)$ the brane spinors of $\psi(x, y = 0)$:

- **Neutrino (Yukawa) portal:** $-\sum_{i,s}\big[y^L_{is}\,\bar{L}_i\tilde{\Phi}P_R\psi^{(L)}_s + y^R_{is}\,\bar{L}_i\tilde{\Phi}P_R\psi^{(R)}_s\big]$ + h.c., where $\psi^{(L)}_s$ ($\psi^{(R)}_s$), $s = 1, 2$, are the brane-right-handed parts of the left (right) half of dirac16complex and $P_R = \frac{1}{2}(1+\gamma_5)$. After symmetry breaking it is a Dirac mass $\mu_{ia} = y_{ia}v/(\sqrt{2}\sqrt{V_\perp})$ between $\nu_{L,i}$ and every Kaluza–Klein mode ($1/\sqrt{V_\perp}$ normalizes the modes on the torus).
- **Vector (current) portal:** $-G^f_P\,(\bar{\psi}\Gamma^\mu\psi)|_W\,(\bar{f}\gamma_\mu f)$ for Standard Model fermions f. In a background of dirac16complex with brane density n_ψ it adds the potential $V^f_P = G^f_P n_\psi$ to the forward scattering of f (for neutrinos: a non-standard matter effect).
- **Higgs (scalar) portal:** $-\lambda_P\,\Phi^\dagger\Phi\,(\bar{\psi}\psi)|_W$, which shifts the dirac16complex mass by $\lambda_P v^2/(2V_\perp)$ after symmetry breaking.

The coupled field equations of all fields:

$$\sum_k \alpha_k E^{(k)\mu}{}_\nu = \kappa\Big(T_\psi + T_{\rm stab} + \frac{\delta^4(y)}{b^4}T_W\Big)^\mu{}_\nu,$$

$$\big(i\Gamma^\mu D_\mu - m\big)\psi - \sum_r g_r \sum_{|K|=r}(\bar{\psi}\Theta_K\psi)\Theta^K\psi = \frac{\delta^4(y)}{b^4}\,J_{\rm portal},$$

with $E^{(k)}$ the Lovelock tensors, T_W the energy–momentum tensor of the Standard Model fields and portals on the brane, $J_{\rm portal} = \partial\mathcal{L}_{\rm portal}/\partial\bar{\psi}$ (for the Yukawa portal $U\big(\sum_i y^*_{is}\tilde{\Phi}^\dagger L_i \otimes e_s + \dots\big)$), and the Standard Model field equations of section 4 with the portal terms as additional sources. In the neutrino sector the Yukawa portal gives, in the basis $(\nu_L, \psi_L)\times(\nu_R, \psi_R)$, the Dirac mass matrix of the zero mode

$$\mathcal{M} = \begin{pmatrix} m_D & \mu \\ 0 & m\,1_4 \end{pmatrix}, \qquad m_D = U_{\rm PMNS}\,{\rm diag}(m_1, m_2, m_3),$$

and for $m \gg \mu$ the light mixing matrix $N = (1-\eta)U_{\rm PMNS}$ is no longer unitary: $1 - NN^\dagger = 2\eta = \mu\mu^\dagger/m^2$ to second order, while the heavy states are mostly dirac16complex. Example: $m = 1$ GeV and μ (in eV)

$$\mu = \begin{pmatrix} 3 & 1 & 0 & 0 \\ 1 & 2 & 0 & 0 \\ -2 & 1 & 0 & 0 \end{pmatrix} \times 10^4 :$$

$$\text{light masses} = (0, 0.0086544743, 0.050338833)\ \text{eV},$$
$$\text{heavy masses} = (1.00000000000, 1.00000000000, 1.00000000025, 1.00000000075)\ \text{GeV},$$
$$(1 - NN^\dagger)_{ee} = 1.00000 \times 10^{-9}.$$

11. Neutrino flavour and flavour conversion (section 19)

A neutrino of flavour α produced with energy E is the superposition $\sum_i U^*_{\alpha i}|\nu_i\rangle$. In vacuum (with $\hbar c = 1.973269804\times10^{-16}$ GeV m) the amplitude after a distance L is $A(\alpha\to\beta)=\sum_i U_{\beta i}U^*_{\alpha i}e^{-im_i^2L/(2E)}$, so

$$P(\alpha\to\beta)=\delta_{\alpha\beta}-4\sum_{i>j}\mathrm{Re}\,X_{ij}\sin^2\Delta_{ij}+2\sum_{i>j}\mathrm{Im}\,X_{ij}\sin 2\Delta_{ij},\qquad X_{ij}=U^*_{\alpha i}U_{\beta i}U_{\alpha j}U^*_{\beta j},\quad \Delta_{ij}=\frac{\Delta m^2_{ij}L}{4E}.$$

In matter of electron density n_e the charged-current forward scattering adds $V_{CC}=\sqrt{2}G_F n_e$ to the ν_e entry of the flavour Hamiltonian ($V_{CC}\to -V_{CC}$ and $U\to U^*$ for antineutrinos; a potential common to all flavours drops out):

$$H=\frac{1}{2E}\,U\,\mathrm{diag}(0,\Delta m^2_{21},\Delta m^2_{31})\,U^\dagger+\mathrm{diag}(V_{CC},0,0),\qquad V_{CC}\big(\rho Y_e=1\ \mathrm{g/cm^3}\big)=7.63\times10^{-14}\ \mathrm{eV}.$$

In two flavours the mixing angle in matter obeys

$$\tan 2\theta_m=\frac{\sin 2\theta}{\cos 2\theta-2EV/\Delta m^2},$$

maximal at the MSW resonance $2EV=\Delta m^2\cos 2\theta$. **Long baseline** (DUNE: $L=1300$ km, $E=2.5$ GeV, $\rho=2.848$ g/cm³, $Y_e=0.5$):

	$P(\nu_\mu\to\nu_e)$	$P(\bar\nu_\mu\to\bar\nu_e)$	$P(\nu_\mu\to\nu_\mu)$
vacuum	0.048836853	0.050198510	0.012813139
matter	0.072326938	0.029191799	—
matter + vector portal $V_{P,e}=0.2V_{CC}$	0.077049983	—	—

The Sun. With the electron density $n_e(r)=245\,N_A\,e^{-10.54\,r/R_\odot}$ cm^{-3} ($R_\odot=6.957\times10^8$ m, N_A Avogadro's number) the two-flavour evolution (effective potential $c^2_{13}V_{CC}$) is integrated from the centre to the surface; the result agrees with the adiabatic MSW formula

$$P_{ee}=c^4_{13}\,\frac{1+\cos 2\theta_{12}\cos 2\theta_m(0)}{2}+s^4_{13},\qquad \text{vacuum average: } c^4_{13}\Big(1-\frac{\sin^2 2\theta_{12}}{2}\Big)+s^4_{13},$$

E **(MeV)**	P_{ee} **(numerical)**	P_{ee} **(adiabatic formula)**
0.300	0.52516444	0.52516451
1.00	0.45841194	0.45841225
5.00	0.31033778	0.31033351
10.0	0.29792183	0.29791749

(at 0.3 MeV the residual matter effect at the centre, $2EV/\Delta m^2_{21}\approx0.15$, still lowers P_{ee} below the vacuum average by about 0.025).

Supernovae. The MSW resonance densities $\rho_{\rm res}=\Delta m^2\cos 2\theta/(2EV_{CC}(1\,\mathrm{g/cm^3}))$ for $E=20$ MeV, $Y_e=0.5$ are 1587.14 g/cm³ (H: Δm^2_{31}, θ_{13}) and 18.9397 g/cm³ (L: Δm^2_{21}, θ_{12}).

Cosmic neutrino background. $T_\nu=(4/11)^{1/3}T_\gamma$ with $T_\gamma=2.7255$ K gives $T_\nu=1.94537$ K and the number density per flavour (neutrinos and antineutrinos) $n_\nu=\frac{3\zeta(3)}{2\pi^2}T_\nu^3=112.016$ cm^{-3}.

dirac16complex and neutrino flavour. With the Yukawa portal the oscillation probabilities of the light states, computed with N in place of U, have $\sum_\beta P(\alpha\to\beta)<1$ (a zero-distance effect, compensated by the heavy states); for $\mu\to0$ they become the three-flavour vacuum probabilities above. The vector portal acts as a non-standard matter potential: $V_{P,e}=0.2\,V_{CC}$ changes $P(\nu_\mu\to\nu_e)$ at DUNE from 0.0723269 to 0.0770500.

12. Verification

Every statement above is a check of 7+1.nb (Wolfram Language 15.0.1) and of the independent Python program tools/ref_sm.py (numpy, scipy, sympy). The numbers printed by both programs:

quantity	values	max. relative difference	note
$v, g, g', \sin^2\theta_W, e, 1/\alpha, \lambda$	7	2.1×10^{-16}	
$\lvert V_{\rm CKM}\rvert$	9	2.3×10^{-5}	Python prints 6 decimals
$J_{\rm CKM}$	1	2.2×10^{-16}	
$\bar\rho, \bar\eta$	2	3.5×10^{-16}	
$\lvert U_{\rm PMNS}\rvert$	9	3.2×10^{-5}	Python prints 5 decimals
$J_{\rm PMNS}$	1	1.2×10^{-16}	
m_1, m_2, m_3 (eV)	3	0	
y_t	1	1.1×10^{-16}	
age (Gyr)	1	1.0×10^{-15}	
$V_\perp$, $V_\perp^{1/4}$, lowest KK mass	3	2.2×10^{-16}	
light (eV) and heavy (GeV) masses with the portal	7	9.9×10^{-7}	
$(1-NN^\dagger)_{ee}$	1	3.3×10^{-7}	
DUNE vacuum $P(\mu\to e)$, $P(\bar\mu\to\bar e)$, $P(\mu\to\mu)$	3	4.3×10^{-16}	
DUNE matter $P(\mu\to e)$, $P(\bar\mu\to\bar e)$	2	1.2×10^{-15}	
vector portal: $P(\mu\to e)$ with, without	2	9.0×10^{-16}	
Sun P_{ee} (E, numerical, adiabatic)	12	3.8×10^{-6}	
supernova resonance densities H, L	2	1.9×10^{-16}	
T_ν, n_ν	2	1.3×10^{-16}	

Apart from the matrices that the Python program prints rounded, the largest relative difference is 3.8×10^{-6}.

The checks of sections 17–19 of the notebook:

#	check	result	section
1	anomaly cancellation per generation (with nu_R): SU(3)^2U(1), SU(2)^2U(1), U(1)^3, grav^2U(1), SU(3)^3 all vanish; doublets per generation even	True	17.1
2	electric charges Q = T3 + Y: (u, d) = (2/3, -1/3), (nu, e) = (0, -1), u_R 2/3, d_R -1/3, e_R -1, nu_R 0	True	17.1
3	g* = 28 + (7/8) 90 = 106.75 for the Standard Model fields (without nu_R)	True	17.1
4	gauge-boson mass matrix: eigenvalues g^2 v^2/4 (twice), (g^2 + g'^2) v^2/4 and 0 (photon)	True	17.3
5	tree level: m_W == g v/2, m_Z == Sqrt[g^2 + g'^2] v/2, rho == m_W^2/(m_Z^2 cos^2 theta_W) == 1, m_H^2 == 2 lambda v^2	True	17.3
6	CKM: unitary; Jarlskog invariant J = 3.12 x 10^-5 (PDG 2024: 3.12 +0.13 -0.12 x 10^-5); A, rhobar, etabar within the PDG uncertainties	True	17.4
7	Yukawa matrices: singular values x v/Sqrt[2] are the quark masses and |U_uL^dag.U_dL| == |V_CKM|	True	17.4
8	PMNS unitary; Dirac neutrino Yukawa Sqrt[2] U.diag(m)/v reproduces the masses (singular values)	True	17.4
9	age of the universe from the brane Friedmann equation: 13.79 Gyr (Planck 2018: 13.787 +- 0.020 Gyr)	True	17.5
10	ADD relation: kappa8 rho4/V_perp == 8 Pi G4 rho4 with G4 = G8/V_perp (kappa8 = 8 Pi G8)	True	17.5
11	brane representation: R_A R_B + R_B R_A == 2 eta17_AB ID16 for R = (gamma_mu (x) 1_4, gamma5 (x) rho_k)	True	18.
12	intertwiner: Gamma17[A].U == U.R_A for A = 0..7, and U is unitary	True	18.
13	chi17 == U.(gamma5 (x) rho4567).U^dag; rho4567 Hermitian with eigenvalues {-1, -1, 1, 1}	True	18.
14	bilinears at the brane: psibar psi' == Sum_a psibar_a psi'_a and psibar Gamma^mu psi' == Sum_a psibar_a gamma^mu psi'_a (mu = 0..3)	True	18.
15	KK operator: X.X == -|q|^2/b0^2 ID16 and X anticommutes with gamma_mu (x) 1_4	True	18.1
16	KK masses: Det[gamma.p (x) 1 + X - m] vanishes on p0^2 - |p|^2 == m^2 + |q|^2/b0^2 (40-digit arithmetic) and not off shell	True	18.1
17	portal mixing: 1 - N.N^dag == mu.mu^dag/m^2 to second order (non-unitary light mixing)	True	18.2

#	check	result	section
18	vacuum: P(a->b) doubly stochastic; equals delta - 4 Sum Re X sin^2 Delta + 2 Sum Im X sin 2Delta; CPT P(a->b) == P(bbar->abar)	True	19.
19	V_CC(rho Y_e = 1 g/cm^3) == 7.63 x 10^-14 eV; matter evolution doubly stochastic; a flavour-universal potential has no effect	True	19.
20	two-flavour MSW: sin^2 theta_m from diagonalization == formula, and sin^2 2theta_m == 1 at 2 E V == dm^2 cos 2theta	True	19.
21	Sun: numerical P_ee == adiabatic MSW formula to 2 x 10^-3 at 0.3, 1, 5 and 10 MeV; P_ee(0.3 MeV) within 0.03 of the vacuum average; P_ee falls with energy	True	19.
22	cosmic neutrino background: T_nu == 1.945 K and n == 112 cm^-3 per flavour	True	19.
23	dirac16complex portal: non-unitary oscillations, Sum_b P(a->b) < 1 (zero-distance effect); decoupled limit == three-flavour vacuum result	True	19.

The Python program's boolean checks:

#	check	result
1	anomaly cancellation in each generation (with nu_R): SU(3)^2U(1), SU(2)^2U(1), U(1)^3, grav^2U(1), SU(3)^3 all 0	True
2	Witten SU(2) anomaly: number of doublets per generation (3 colours of Q + 1 L) = 4, even	True
3	electric charges Q = T3 + Y: Q -> (2/3, -1/3), u_R 2/3, d_R -1/3, L -> (0, -1), e_R -1, nu_R 0	True
4	effective number of relativistic degrees of freedom g* = 28 + (7/8) 90 = 106.75 (106.75 + 5.25 if nu_R were thermal)	True
5	gauge-boson mass matrix from |D<Phi>|^2: eigenvalues g^2v^2/4 (twice, W+-), (g^2+g'^2)v^2/4 (Z), 0 (photon)	True
6	tree-level relations: m_W = g v/2, m_Z = sqrt(g^2+g'^2) v/2, rho = m_W^2/(m_Z^2 cos^2) = 1, m_H^2 = 2 lambda v^2	True
7	CKM unitarity V V^dag = 1	True
8	J = c12 c23 c13^2 s12 s23 s13 sin(delta) = 3.12e-5 (PDG 2024: 3.12 +0.13 -0.12 e-5)	True
9	Wolfenstein parameters agree with the PDG fit (A 0.826, rhobar 0.1591, etabar 0.3523) within their quoted uncertainties	True
10	Yukawa matrices Y_u = sqrt2 diag(m_u,m_c,m_t)/v, Y_d = sqrt2 V diag(m_d,m_s,m_b)/v: singular values x v/sqrt2 = masses, |U_uL^dag U_dL| = |V|	True
11	PMNS unitarity	True
12	neutrino Yukawa matrix Y_nu = sqrt2 U diag(m1,m2,m3)/v: masses and |U| recovered (singular value decomposition)	True
13	vacuum: probabilities P(a -> b) form a doubly stochastic matrix (rows and columns sum to 1)	True
14	vacuum: amplitude result = delta_ab - 4 sum_{i>j} Re(X_ij) sin^2 Delta_ij + 2 sum_{i>j} Im(X_ij) sin 2Delta_ij, X_ij = U_ai* U_bi U_aj U_bj*	True
15	CPT: P(a->b) = P(bbar->abar); CP asymmetry P(mu->e) - P(mubar->ebar) proportional to J (vanishes for delta = 0, pi)	True
16	V_CC(rho Y_e = 1 g/cm^3) = 7.63e-14 eV	True
17	matter (constant density): unitary evolution, probabilities doubly stochastic	True
18	matter: a flavour-universal potential has no effect (only differences of potentials matter)	True
19	two-flavour MSW: sin^2 theta_m from diagonalization = formula; maximal mixing (sin^2 2theta_m = 1) at 2 E V = dm^2 cos 2theta	True
20	Sun: numerical P_ee = adiabatic MSW formula to 2e-3 at E = 0.3, 1, 5, 10 MeV	True
21	Sun: P_ee falls from near the vacuum average c13^4 (1 - sin^2(2 theta12)/2) + s13^4 (within 0.03 at 0.3 MeV) to about 0.31 at 10 MeV	True
22	CNB: T_nu = 1.945 K and n = 112 cm^-3 per flavour	True
23	brane representation: gamma_mu (x) 1_4 (mu = 0..3) and gamma5 (x) rho_k (k = 4..7) satisfy {R_A, R_B} = 2 eta_AB	True
24	intertwiner U (Schur average over the 256 Clifford products): Gamma17[A] U = U R_A for A = 0..7, U unitary	True
25	8D chirality at the brane: chi17 = U (gamma5 (x) rho4567) U^dag, rho4567 Hermitian with eigenvalues +1, +1, -1, -1	True
26	bilinears at the brane: Psibar Psi' = sum_a psibar_a psi'_a and Psibar Gamma^mu Psi' = sum_a psibar_a gamma^mu psi'_a (mu = 0..3)	True
27	Kaluza-Klein: the 16x16 operator gamma.p (x) 1 + gamma5 (x) rho.q/b0 - m is singular exactly on p0^2 - |p|^2 = m^2 + |q|^2/b0^2	True
28	KK operator: X^2 = -|q|^2/b0^2 and X anticommutes with gamma^mu (x) 1	True
29	non-unitary oscillations: sum_b P(a->b) = (N N^dag)_aa^2-type deficit < 1 (zero-distance effect), complemented by the heavy states	True
30	portal decoupling: for mu -> 0 the light block becomes the PMNS matrix and the oscillation probabilities the three-flavour vacuum ones (section 19)	True
31	age 13.79 Gyr (Planck 2018: 13.787 +- 0.020)	True

#	check	result
32	`ADD relation: 3 Ha^2 = kappa8 rho4/V_perp = 8 pi G4 rho4 with G4 = G8/V_perp`	True

13. How to reproduce

1. `python3 tools/make_7p1_nb.py` writes 7+1.nb (unevaluated).
2. Evaluate 7+1.nb in a fresh kernel of Wolfram 15.0.1 (Evaluation ▷ Quit Kernel, then Evaluation ▷ Evaluate Notebook; or `Get["D:/Developer/github/7+1/tools/run_eval.wl"]`) and save it; the last section summarizes all checks.
3. `python3 tools/ref_sm.py` repeats the checks of this document independently (exit code 0 if all pass).
4. `python3 tools/make_docs.py` regenerates this document, its symbol dictionary and its provenance record.

14. References

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